

Solar Fe I appendix

Sun · Fe I · downloadable appendix

Fe.json v1.163 · 77 Fe I products · 92 total

Solar Fe I anchor: 7.477 ± 0.052

HARPS · Reference Grade · Amarsi 3D-NLTE · $n = 21$. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.554 ± 0.131

Kitt Peak · Synth · 1D-LTE · Deep Grade · $n = 55$

VIS

7.504 ± 0.049

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade · $n = 21$

VIS

7.477 ± 0.052

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade · $n = 21$

red-optical

7.498 ± 0.059

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · $n = 69$

NIR

7.551 ± 0.087

CRIRES+ · Synth · 1D-LTE · Codex Grade · $n = 5$

H

7.523 ± 0.084

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · $n = 19$

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Solid bars show statistical uncertainty from the line measurements; wireframes show systematic uncertainty from the product's adopted error budget, which does not average away with more lines. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

Reference Grade. An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

Codex Grade. The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.

Deep Grade. The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

3D-NLTE (Amarsi) includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

Mean-3D / Frankenstein combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

Synthesis fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See [Methodology](#) for the full model, line-selection and uncertainty procedures.

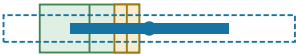
Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

blue solid = statistical σ · wireframe = systematic σ · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders · N/A = no product for this model

near-UV — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.36-7.71

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=55		7.554 ± 0.061 stat ± 0.116 syst
Synth · 1D-LTE · Gerber no product		N/A
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Deep Grade · n=55		7.562 ± 0.061 stat ± 0.116 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

near-UV — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band) A(Fe) dex; axis 7.36-7.71

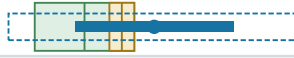
EW · 1D-LTE no product		N/A
Synth · 1D-LTE Deep Grade · n=54		7.508 ± 0.064 stat ± 0.117 syst
Synth · 1D-LTE · Gerber no product		N/A

**near-UV — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.36-7.71**

EW · 1D-NLTE · Bergemann
no product

N/A

Synth · 1D-NLTE · Bergemann
Deep Grade · n=54



7.516
±0.064 stat ±0.117 syst

Synth · 1D-NLTE · Gerber
no product

N/A

Synth · <3D>-LTE · Gerber · stagger
no product

N/A

Synth · <3D>-NLTE · Gerber · stagger
no product

N/A

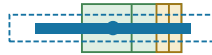
Synth · 3D-NLTE · Amarsi · lab-gf
no product

N/A

near-UV axis: 7.36 · 7.45 · 7.53 · 7.62 · 7.71 A(Fe) dex

**VIS — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.31-7.66**

EW · 1D-LTE
Codex Grade · n=13 · 1 alternate



7.445
±0.062 stat ±0.083 syst

Synth · 1D-LTE
Codex Grade · n=67



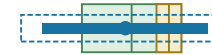
7.442
±0.019 stat ±0.081 syst

Synth · 1D-LTE · Gerber
Codex Grade · n=67



7.439
±0.021 stat ±0.063 syst

EW · 1D-NLTE · Bergemann
Codex Grade · n=12



7.455
±0.067 stat ±0.084 syst

Synth · 1D-NLTE · Bergemann
Codex Grade · n=61



7.448
±0.019 stat ±0.081 syst

Synth · 1D-NLTE · Gerber
Codex Grade · n=67



7.488
±0.020 stat ±0.063 syst

Synth · <3D>-LTE · Gerber · stagger · Frankenstein
(experimental)
Codex Grade · n=67



7.541
±0.020 stat ±0.063 syst

Synth · <3D>-NLTE · Gerber · stagger · Frankenstein
(experimental)
Codex Grade · n=67



7.562
±0.020 stat ±0.063 syst

Synth · 3D-NLTE · Amarsi · lab-gf
Codex Grade · n=50



7.505
±0.022 stat ±0.058 syst

**VIS — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.31-7.66**

EW · 1D-LTE
Codex Grade · n=13



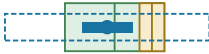
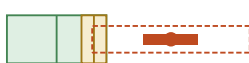
7.463
±0.062 stat ±0.092 syst

Synth · 1D-LTE
Codex Grade · n=67

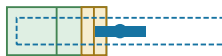


7.447
±0.020 stat ±0.081 syst

VIS — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected A(Fe) dex; axis 7.31-7.66

Synth · 1D-LTE · Gerber Codex Grade · n=67		7.451 ±0.021 stat ±0.063 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.464 ±0.066 stat ±0.091 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=61		7.454 ±0.020 stat ±0.082 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=67		7.497 ±0.022 stat ±0.063 syst
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.552 ±0.022 stat ±0.063 syst
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.552 ±0.022 stat ±0.063 syst
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.022 stat ±0.058 syst

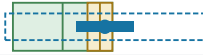
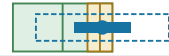
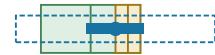
VIS — HARPS solar_harps_molecfit_corrected A(Fe) dex; axis 7.31-7.66

EW · 1D-LTE Codex Grade · n=6 · 1 alternate		7.498 ±0.070 stat ±0.097 syst
Synth · 1D-LTE Codex Grade · n=67		7.511 ±0.020 stat ±0.083 syst
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.509 ±0.021 stat ±0.063 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=5		7.433 ±0.077 stat ±0.095 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=61		7.504 ±0.020 stat ±0.083 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=67		7.545 ±0.020 stat ±0.063 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.024 stat ±0.058 syst

VIS — IAG solar_iag A(Fe) dex; axis 7.31-7.66

EW · 1D-LTE Codex Grade · n=5		7.498 ±0.044 stat ±0.093 syst
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VIS — IAG solar_iag A(Fe) dex; axis 7.31-7.66

Synth · 1D-LTE Codex Grade · n=43		7.494 ±0.023 stat ±0.080 syst
Synth · 1D-LTE · Gerber Codex Grade · n=43		7.492 ±0.023 stat ±0.053 syst
EW · 1D-NLTE · Bergemann Codex Grade · n=4		7.481 ±0.019 stat ±0.089 syst
Synth · 1D-NLTE · Bergemann Codex Grade · n=37		7.480 ±0.023 stat ±0.080 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=43		7.549 ±0.023 stat ±0.053 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=39		7.543 ±0.025 stat ±0.056 syst

VIS axis: 7.31 · 7.39 · 7.48 · 7.57 · 7.66 A(Fe) dex

red-optical — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected A(Fe) dex; axis 7.35-7.59

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=69		7.458 ±0.017 stat ±0.069 syst
Synth · 1D-LTE · Gerber Codex Grade · n=69		7.454 ±0.016 stat ±0.057 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=35		7.451 ±0.017 stat ±0.071 syst
Synth · 1D-NLTE · Gerber Codex Grade · n=69		7.498 ±0.016 stat ±0.057 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

red-optical — Kitt Peak — 1984 Molecfiit corrected solar_kpno_molecfiit_corrected A(Fe) dex; axis 7.35-7.59

EW · 1D-LTE
no product

N/A

Synth · 1D-LTE
Codex Grade · n=53



7.484
±0.029 stat ±0.070 syst

Synth · 1D-LTE · Gerber
Codex Grade · n=53



7.478
±0.028 stat ±0.055 syst

EW · 1D-NLTE · Bergemann
no product

N/A

Synth · 1D-NLTE · Bergemann
Codex Grade · n=27



7.493
±0.025 stat ±0.070 syst

Synth · 1D-NLTE · Gerber
Codex Grade · n=53



7.507
±0.029 stat ±0.055 syst

Synth · <3D>-LTE · Gerber · stagger
no product

N/A

Synth · <3D>-NLTE · Gerber · stagger
no product

N/A

Synth · 3D-NLTE · Amarsi · lab-gf
no product

N/A

red-optical — IAG solar_iag A(Fe) dex; axis 7.35-7.59

EW · 1D-LTE
no product

N/A

Synth · 1D-LTE
Codex Grade · n=68



7.448
±0.024 stat ±0.070 syst

Synth · 1D-LTE · Gerber
Codex Grade · n=68



7.453
±0.024 stat ±0.055 syst

EW · 1D-NLTE · Bergemann
no product

N/A

Synth · 1D-NLTE · Bergemann
Codex Grade · n=35



7.460
±0.041 stat ±0.074 syst

Synth · 1D-NLTE · Gerber
Codex Grade · n=68



7.491
±0.024 stat ±0.055 syst

Synth · <3D>-LTE · Gerber · stagger
no product

N/A

Synth · <3D>-NLTE · Gerber · stagger
no product

N/A

Synth · 3D-NLTE · Amarsi · lab-gf
no product

N/A

red-optical axis: 7.35 · 7.41 · 7.47 · 7.53 · 7.59 A(Fe) dex

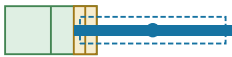
NIR — Kitt Peak — Kurucz 2005 corrected solar_kpno_kurucz2005_corrected A(Fe) dex; axis 7.36-7.73

EW · 1D-LTE no product		N/A
Synth · 1D-LTE no product		N/A
Synth · 1D-LTE · Gerber Codex Grade · n=23		7.515 ±0.086 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

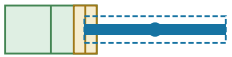
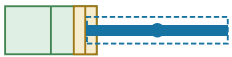
NIR — Kitt Peak — 1984 Molecfit corrected solar_kpno_molecfit_corrected A(Fe) dex; axis 7.36-7.73

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=23		7.607 ±0.080 stat ±0.065 syst
Synth · 1D-LTE · Gerber Codex Grade · n=23		7.601 ±0.080 stat ±0.063 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=6		7.488 ±0.059 stat ±0.065 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — IAG solar_iag A(Fe) dex; axis 7.36-7.73

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=25		7.546 ±0.159 stat ±0.063 syst
Synth · 1D-LTE · Gerber Codex Grade · n=22		7.549 ±0.069 stat ±0.064 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=6		7.599 ±0.072 stat ±0.063 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR — CRIRES+ solar_crires_plus_y_wide_rya1054 A(Fe) dex; axis 7.36-7.73

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=5		7.551 ±0.062 stat ±0.062 syst
Synth · 1D-LTE · Gerber Codex Grade · n=5		7.553 ±0.062 stat ±0.061 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann Codex Grade · n=2		7.492 ±0.070 stat ±0.061 syst
Synth · 1D-NLTE · Gerber no product		N/A
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

NIR axis: 7.36 · 7.45 · 7.55 · 7.64 · 7.73 A(Fe) dex

H — CRIRES+ solar_crires_plus_h_rya1094 A(Fe) dex; axis 7.39-7.67

EW · 1D-LTE no product		N/A
Synth · 1D-LTE Codex Grade · n=20		7.552 ±0.087 stat ±0.072 syst
Synth · 1D-LTE · Gerber Codex Grade · n=19		7.523 ±0.051 stat ±0.067 syst
EW · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Bergemann no product		N/A
Synth · 1D-NLTE · Gerber Codex Grade · n=9		7.549 ±0.070 stat ±0.065 syst
Synth · <3D>-LTE · Gerber · stagger no product		N/A
Synth · <3D>-NLTE · Gerber · stagger no product		N/A
Synth · 3D-NLTE · Amarsi · lab-gf no product		N/A

H axis: 7.39 · 7.46 · 7.53 · 7.60 · 7.67 A(Fe) dex

Read all product values, including alternate line sets and experiments

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_h_rya1094 · GRADED · SYNTH · 1D-LTE: 7.552 ± 0.113 reported total; n=20; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · solar_crires_plus_h_rya1094 · GRADED · SYNTH · ENGINE-B-NLTE: 7.549 ± 0.096 reported total; n=9; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_h_rya1094 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.523 ± 0.084 reported total; n=19; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · 1D-LTE: 7.551 ± 0.087 reported total; n=5; ξ MEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · ENGINE-A: 7.492 ± 0.093 reported total; n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.553 ± 0.087 reported total; n=5; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.546 ± 0.171 reported total; n=25; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.599 ± 0.095 reported total; n=6; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.549 ± 0.094 reported total; n=22; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent

from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.515 ± 0.105 reported total; n=23; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfits_corrected · GRADED · SYNTH · 1D-LTE: 7.607 ± 0.104 reported total; n=23; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfits_corrected · GRADED · SYNTH · ENGINE-A: 7.488 ± 0.087 reported total; n=6; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfits_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.601 ± 0.102 reported total; n=23; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfits_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.477 ± 0.052 reported total; n=21; ξ NOT_APPLICABLE

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED-LOCALRENORM · SYNTH · 1D-LTE: 7.339 ± 0.087 reported total; n=109; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED-LOCALRENORM · SYNTH · ENGINE-A: 7.350 ± 0.088 reported total; n=105; ξ MEASURED

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.535 ± 0.093 reported total; n=109; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfits_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.554 ± 0.094 reported total; n=105; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfits_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.120 reported total; n=6; ξ MEASURED

HARPS · EW · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfits_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.433 ± 0.123 reported total; n=5; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfits_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.431 ± 0.122 reported total; n=5; ξ MEASURED

HARPS · Synth · 1D-LTE · Codex Grade · solar_harps_molecfits_corrected · GRADED · SYNTH · 1D-LTE: 7.511 ± 0.086 reported total; n=67; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_harps_molecfits_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfits_corrected · GRADED · SYNTH · ENGINE-A: 7.504 ± 0.086 reported total; n=61; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Codex Grade · solar_harps_molecfits_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.545 ± 0.067 reported total; n=67; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

HARPS · Synth · 1D-LTE · Gerber · Codex Grade · solar_harps_molecfits_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.509 ± 0.067 reported total; n=67; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_iag · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.050 reported total; n=21; ξ NOT_APPLICABLE

IAG · Synth · 1D-LTE · Deep Grade · solar_iag · DEEPGRADED · SYNTH · 1D-LTE: 7.416 ± 0.071 reported total; n=41; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_iag · DEEPGRADED · SYNTH · ENGINE-A: 7.422 ± 0.070 reported total; n=37; ξ MEASURED

IAG · EW · 1D-LTE · Codex Grade · solar_iag · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.102 reported total; n=5; ξ MEASURED

IAG · EW · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · PROFILEFIT · ENGINE-A: 7.481 ± 0.091 reported total; n=4; ξ MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.494 ± 0.083 reported total; n=43; ξ MEASURED

IAG · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.543 ± 0.061 reported total; n=39; ξ NOT_APPLICABLE

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.480 ± 0.083 reported total; n=37; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.549 ± 0.058 reported total; n=43; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.492 ± 0.058 reported total; n=43; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.471 ± 0.049 reported total; n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.421 ± 0.086 reported total; n=108; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.431 ± 0.086 reported total; n=104; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.445 ± 0.104 reported total; n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.455 ± 0.107 reported total; n=12; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.479 ± 0.090 reported total; n=9; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.442 ± 0.084 reported total; n=67; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.505 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.448 ± 0.084 reported total; n=61; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.488 ± 0.066 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.439 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.541 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.066 reported total; n=67; ξ UNMEASURED ·

INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfite_corrected · ASPLUND_AGSS21 · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.049 reported total; n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfite_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.417 ± 0.045 reported total; n=108; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfite_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.426 ± 0.084 reported total; n=104; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_molecfite_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.463 ± 0.111 reported total; n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfite_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.464 ± 0.112 reported total; n=12; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · 1D-LTE: 7.447 ± 0.084 reported total; n=67; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.062 reported total; n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · ENGINE-A: 7.454 ± 0.085 reported total; n=61; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.497 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.451 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.552 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_molecfite_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.552 ± 0.067 reported total; n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.554 ± 0.131 reported total; n=55; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.562 ± 0.131 reported total; n=55; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfite_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.508 ± 0.133 reported total; n=54; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfite_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.516 ± 0.133 reported total; n=54; ξ MEASURED

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.448 ± 0.074 reported total; n=68; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.460 ± 0.084 reported total; n=35; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.491 ± 0.060 reported total; n=68; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent

from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.453 ± 0.060 reported total; n=68; ξ NOT_IN_CAMPAIGN · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.458 ± 0.071 reported total; n=69; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.451 ± 0.073 reported total; n=35; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.498 ± 0.059 reported total; n=69; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.454 ± 0.059 reported total; n=69; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.484 ± 0.075 reported total; n=53; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.493 ± 0.075 reported total; n=27; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.507 ± 0.062 reported total; n=53; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.478 ± 0.062 reported total; n=53; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline, which rests on the VIS anchor 7.466 (RYA-161: nothing calibrated to a target).

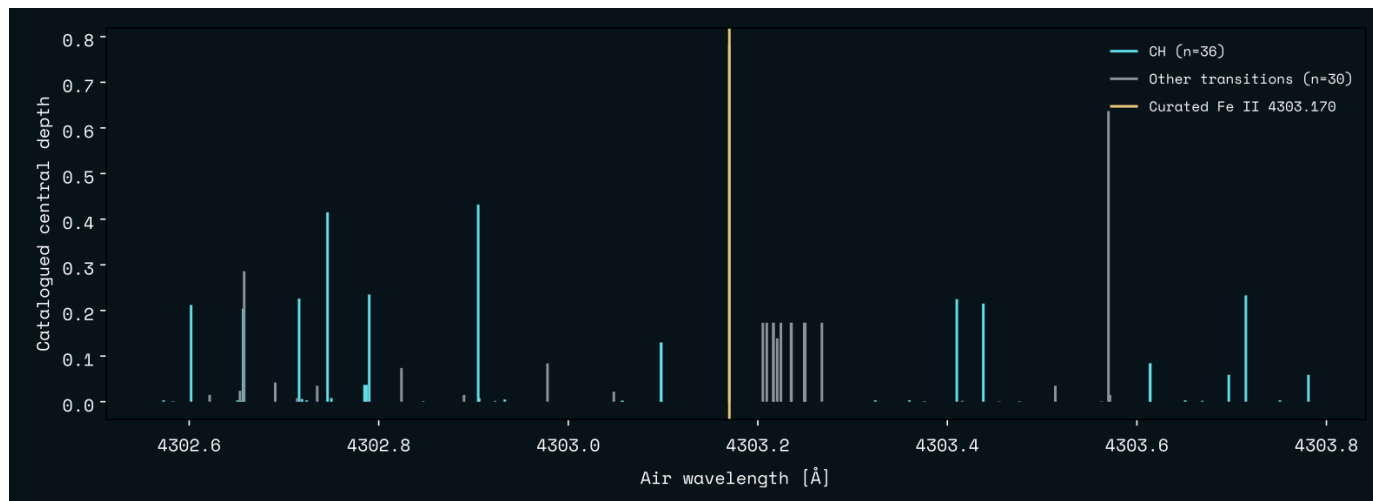
Frankenstein — experimental, pending the Bride

4 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

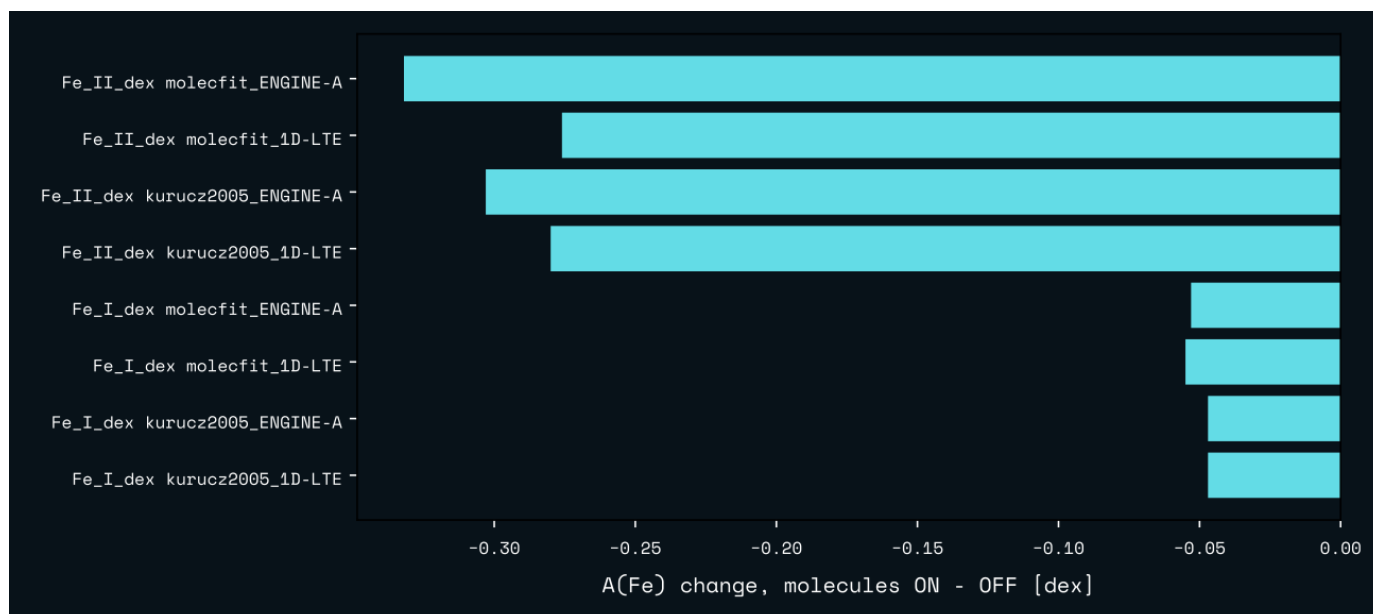
Problem-line and model diagnostics

CH G-band artifact



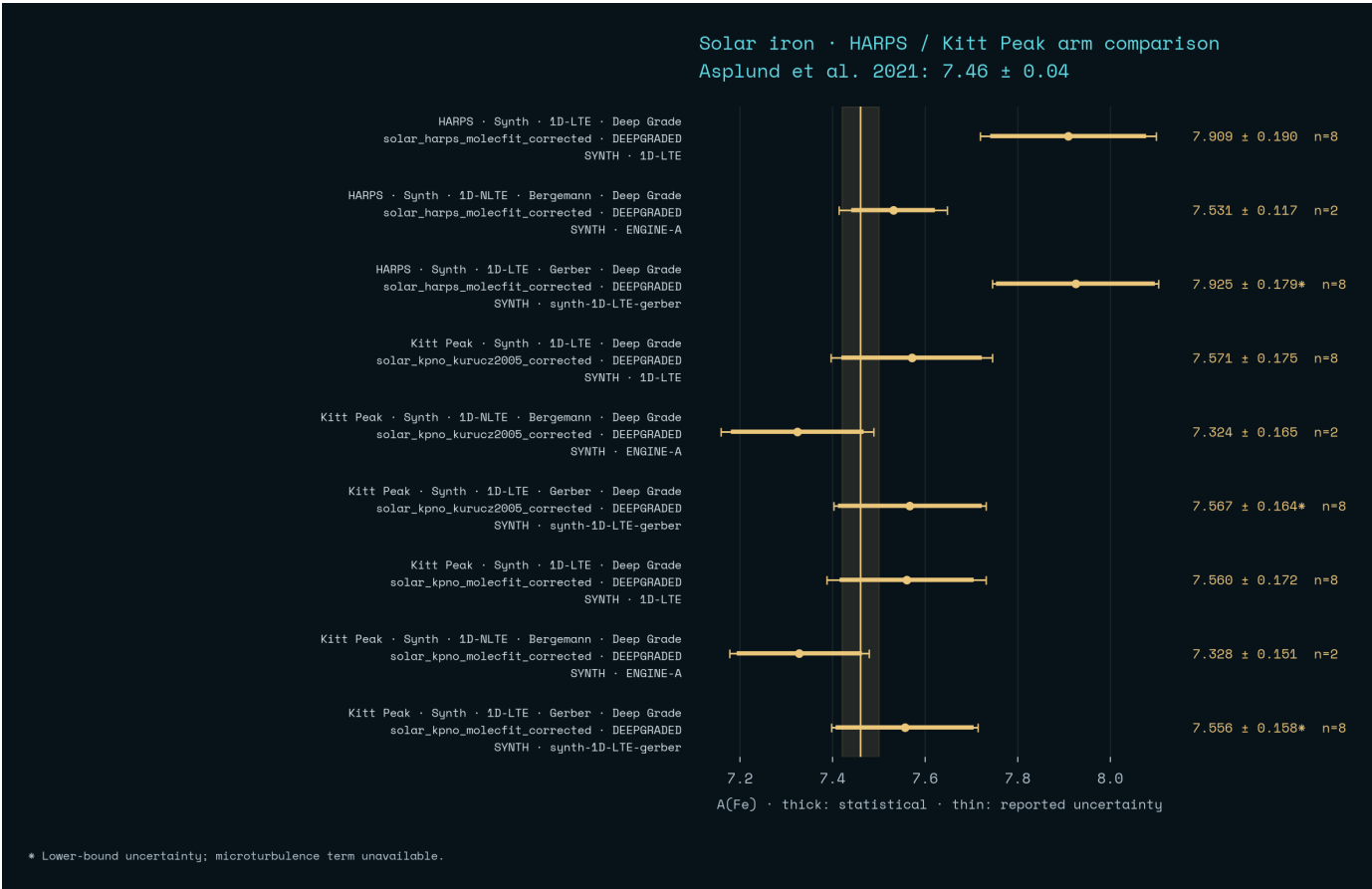
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · $A = 10.0$ (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 Å fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator's fit window; it is not a synthetic spectrum. [Source evidence](#)

Near-UV molecular opacity



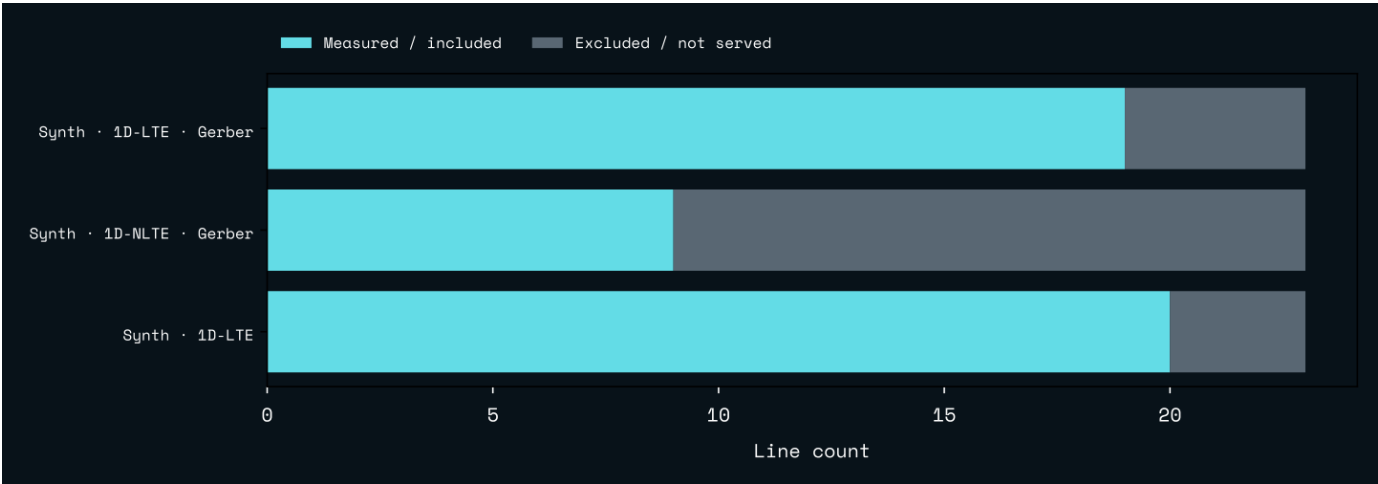
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. [Source evidence](#)

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. [Source evidence](#)

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. [Source evidence](#)

Download the evidence

2676 graded per-line measurements exported. Evidence for 21 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

[Graded per-line abundances \(CSV\)](#)

[All product fields \(CSV; nested fields preserved as JSON\)](#)

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0

Science source commit 370a66e8de93b5b3453edd56114f2814a0d06bed

Fe.json v1.163 · feed timestamp 2026-09-11T04:50:42Z

Generated 2026-09-11T05:30:36+00:00

Feed SHA-256 0b49055ba5dc5931f126b2dba263c387570e6ba5df1893ef57c1bbde33442b62

[Refreshed product tracker](#) · [Build manifest](#) · [Fe II appendix](#)

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Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). [Fe I oscillator strengths for the Gaia-ESO survey](#) (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). [Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels](#) (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.